Coal Dispatch Flow Mine to Railway Siding

DETAILED OPERATIONAL OVERVIEW

Coal Dispatch Flow Mine to Railway Siding

STEP 1: VEHICLE AUTHORIZATION & ENTRY INTO MINE

All vehicles are pre-authorized in the internal system through RFID

Each vehicle is GPS-enabled and monitored 24×7 from the Control Room.

Vehicle is allowed entry only after system validation.

STEP 2: GATE PASS / LOADING SLIP ISSUE

On entry gate, vehicle must **handover the previous trip gate pass** issued at the railway siding.

New gate pass / loading slip is issued **only after verification** of previous trip completion.

Tare Weight is done after obtaining gate-pass at the entry point.
(In some conditions, tare weight may be taken every fort night)

Coal Dispatch Flow Mine to Railway Siding

STEP 3: MOVEMENT TO LOADING POINT

Authorized vehicle proceeds directly to the **coal loading point** after getting a loading slip.

Vehicle stands in **queue as per operational sequence**.

STEP 4: COAL LOADING AT MINE

Coal is loaded by **loader operators** available at loading point.

Operators ensure, No overloading / No Underloading.

Loading is done strictly as per **permitted carrying capacity**.

Coal Dispatch Flow Mine to Railway Siding

STEP 5: WEIGHBRIDGE MOVEMENT & WEIGHMENT

After loading, vehicle moves to the **weighbridge**.

Weighment is carried out and **weighment slip is generated**.

If vehicle overloaded, then again go back to stock yard and drop excess load.
then again come back to weighbridge for obtaining loading slip.

STEP 6: MINING CHALLAN GENERATION (JIMMS)

Based on weighment slip and approved routed, the JIMMS challan is generated.

STEP 7: DUST SUPPRESSION & SAFETY COMPLIANCE

Loaded vehicle passes through **water sprinklers** to suppress dust.

After sprinkling entire vehicle body is **covered with tarpaulin**

Coal Dispatch Flow Mine to Railway Siding

STEP 8: EXIT INSPECTION & VERIFICATION

At mine exit point:

- Weighment slip is verified
- Mining challan is verified

Physical inspection ensures:

- Proper tarpaulin covering
- No spillage risk

STEP 9: EXIT FROM MINE PREMISES

After successful verification vehicle is allowed to **exit the mine.**

Coal Dispatch Flow Mine to Railway Siding

STEP 10: GPS-BASED TRANSIT MONITORING

Vehicle movement is tracked **live through GPS**.

Control Room monitors:

- Route adherence
- Stoppage or deviation
- Speed and delay

STEP 11: ROAD PATROLLING

Dedicated **road patrolling teams** will monitor vehicle movement.

Ensures:

- Timely arrival at railway siding
- No pilferage enroute and no deviations.

Coal Dispatch Flow Mine to Railway Siding

STEP 12: COAL RECEIPT AT RAILWAY SIDING

Loaded trucks directly goes to the weighbridge at railway siding and ensure that same load received.

WB operator updates trips received and quantity received in siding

Siding stock keeps increasing continuously.

STEP 13: UNLOADED TRUCKS WEIGHMENT

After unloading, trucks will directly move to the weighbridge again for tare weight.

This process will ensure that the truck is completely unloaded.

OPERATIONS AT RAILWAY SIDING

STEP 1: INDENT FOR RAKE PLACEMENT

Railway Siding In-Charge checks:

- Available siding stock
- Target rake quantity

Based on stock availability, **indent is raised to Railways** for rake placement.

Important:

Indent is **stock-driven**, not time-driven.

STEP 2: RAKE AVAILABILITY & PLACEMENT BY RAILWAYS

After receiving indent:

- Railways place a rake **based on availability of rake.**

OPERATIONS AT RAILWAY SIDING

STEP 3: TXR INSPECTION (BEFORE LOADING)

Once rake is placed:

- **TXR (Train Examiner) inspection** is carried out
- TXR inspection is conducted by **Railway staff**.

TXR Inspection Includes:

Checking all wagon doors

Checking wagon condition (fit / unfit)

If any wagons found unfit, then a memo will be issued by railway guard for such unfit wagons.

OPERATIONS AT RAILWAY SIDING

STEP 4: MARKING OF UNFIT WAGONS

If TXR declares wagons as unfit:

- Our team marks them clearly:
  - **Night:** Radium reflectors / red lights
  - **Day:** Red flags

These wagons are **excluded from loading**

OPERATIONS AT RAILWAY SIDING

STEP 5: START OF LOADING TIME (CRITICAL STEP)

Loading time starts from rake placement

Loading Rules:

Maximum free loading time: 3 hours

Even **5 minutes extra** beyond 3 hours:

- Railway charges **1 full hour penalty**

STEP 6: WAGON LOADING PROCESS

Loading starts **only after TXR clearance**

Loader operators load wagons **one by one,** and ensure No overloading / No underloading.

- More or less weighment is to be ensured by inbuilt load cells of the loader buckets. **Unfit wagons are skipped**

OPERATIONS AT RAILWAY SIDING

STEP 7: POST-LOADING INSPECTION BY RAILWAY GUARD

After loading completion:

- Message given to **Railway Guard**

Guard inspects:

- Coal spillage on tracks
- Coal between wagons

If Spillage Found; Spillage is removed and cleared AND Track made safe

STEP 8: CLEARANCE FOR MOVEMENT

After clearance:

- Guard informs **Loco Pilot**
- Permission given to move rake

OPERATIONS AT RAILWAY SIDING

STEP 9: IN-MOTION WEIGHMENT (VERY SENSITIVE STEP)

Loco pilot moves rake from:

- Loading track → **In-motion weighment track**

Time Limit

20 minutes allowed

This 20 minute is **included in the same 3-hour free time**

Speed Conditions:

Rake speed must be **5–7 kmph**

OPERATIONS AT RAILWAY SIDING

STEP 9: IN-MOTION WEIGHMENT (VERY SENSITIVE STEP) Contd..

- **These issues can cause:**
  - Wrong weighment
  - Overloading indication

If Weighment Fails:

- Rake is taken forward
- Brought back again on weighment track
- Re-weighment done

OPERATIONS AT RAILWAY SIDING

STEP 10: GENERATION OF COMMERCIAL RR (FINAL & BINDING)

After **successful weighment**:

- **Commercial Railway Receipt (RR)** is generated

RR Contains:

Charged weight

Overloading / underloading

Time taken

Demurrage

Penalties

Freight & GST

RR is the final authority document

OPERATIONS AT RAILWAY SIDING

STEP 11: RECEIPT OF RR & DOCUMENTS

At Railway Siding, we receive:

- In-motion weighment slip
- TXR unfit memo
- Forwarding copy

Note: Railway sends RR **directly to the coal company.**

Impact of underloading wagons:

Railways don’t have any concern on underloading, how ever the coal company will lose the revenue.